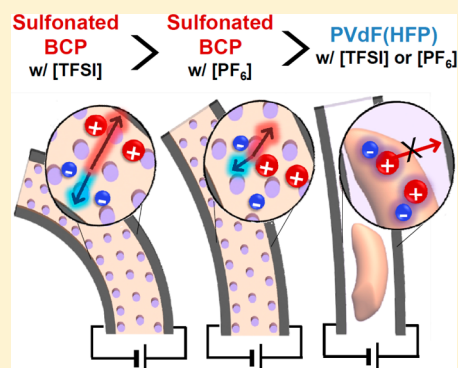


Factors Affecting Electromechanical Properties of Ionic Polymer Actuators Based on Ionic Liquid-Containing Sulfonated Block Copolymers

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ABSTRACT: We investigate the electromechanical properties of ionic polymer actuators based on sulfonated block copolymers and imidazolium ionic liquids (ILs) to unveil key factors affecting the actuation performance. First, the extent of electromechanical deformation of the actuators was proven to be largely affected by the type of anion in the ILs, as understood by the anion-dependent ionic conductivity, charging time, and Young's modulus of IL-containing polymers. In particular, upon switching the anion from hexafluorophosphate to bis-(trifluoromethane)sulfonimide, more than a 2-fold enhancement in the bending strain was observed, which has not been previously reported. Second, we show that the bending strain and durability of the actuators are tunable in a straightforward manner by controlling the block architecture and molecular weight of the polymer, where the use of triblock copolymers was found to be beneficial in enhancing the actuation performance. The high mechanical strength of IL-containing triblock copolymers at large IL loadings was responsible for advancing the actuation performance because of bridge/loop formations in the hydrophobic domains of the triblock copolymers, which is essentially not affected by the incorporation of ILs. Lastly, our actuators demonstrated up to a 10 times increase in the bending strain, compared to actuators based on conventional poly(vinylidene fluoride-co-hexafluoropropene) with the same type of IL. The key to success stemmed from the self-assembled structure of sulfonated block copolymers having continuous ionic phases with highly interactive $-\text{SO}_3\text{H}$ surfaces, which facilitated fast ion diffusion with a reduced tendency to form ion clusters. A molecular-level understanding of actuation mechanisms underlying the improved actuation performance was acquired by combining *in situ* spectroscopy and *in situ* scattering experiments under voltage stimuli.



INTRODUCTION

In recent years, considerable attention has been given to electroactive polymers (EAPs) that show a mechanical response to an applied electric field, aimed at applications of actuators¹ and sensors.² Actuator functionalities of EAPs are particularly fascinating because they can be driven under low-voltage conditions ($<5\text{ V}$),³ paving the way for the development of low-power electrochemical devices such as artificial muscles⁴ and microelectronic modules.⁵ The inherent features of polymers such as flexibility, lightweight, and ease of manufacturing into various shapes have also been highlighted as desirable properties of cutting-edge technologies to be integrated into humanoid robots and microelectromechanical systems.⁸

Ionic polymers are of increasing interest for EAP actuator applications, in which bending motions are caused by ion migration in the direction of the oppositely charged electrode surface under an applied electric field.^{7,8} One of the most commonly employed ionic polymers for EAP actuators is a perfluorosulfonic acid ionomer (commercialized under the trademark Nafion),^{9–12} which has the advantage of high conductivity in the presence of moisture. However, reported actuators based on humidified Nafion are not yet completely developed, and several issues still need to be examined. The

most problematic aspect is that the actuation performance is largely dependent on the degree of hydration, which impedes the long-term use of actuators under ambient conditions.¹⁰ The narrow window of electrochemical stability, slow bending responses in time scales of the order tens of seconds, and uncontrollable motion by back relaxation further discouraged their widespread use in various applications.^{11,12}

Thus, a large number of recent studies have exploited a range of high-boiling plasticizers in actuators, intended to replace water while preserving their high ionic conductivity.^{13,14} This prompted intensive efforts to utilize the polymers containing ionic liquids (ILs) in EAP actuators, as pioneered by Terasawa et al.,^{15–17} to include the fascinating physicochemical properties of ILs characterized by negligible vapor pressures, high ionic conductivities, and wide windows of electrochemical stability. Moreover, hydrophobic ILs contributed to significant progress in the era of EAP actuators by enabling durable operation in air.¹⁸ Given that the electromechanical response of EAP actuators is governed by ion diffusion dynamics, choosing

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Table 1. Molecular Properties of Polymers and Ionic Liquids Used in the Present Study

sample code	polymer used	MW (kg/mol)	SL (mol %)	ϕ_{PSS}^a (vol %)	domain size ^b (nm)	morphology ^b	Young's modulus ^c (GPa)
D17(42)	PSS–PMB	22.3–18.7	42	45	41.9	HEX	2.1
T7(45)	PSS–PMB–PSS	9.5–19.6–9.8	45	40	21.9	HEX	3.2
T12(40)	PSS–PMB–PSS	16.2–25.1–16.8	40	49	27.2	HEX	4.2
ionic liquids used	MW (kg/mol)	V_w^d (Å ³)	T_m^e (°C)	viscosity (cP)	conductivity ^f (mS/cm)	solvation time ⁱ (ps)	binding energy ^{j,k} (kJ/mol)
[HmIm][BF ₄]	167/87	185.9/54.2	−82	314 ^f	1.22	2588	395
[HmIm][PF ₆]	167/145	185.9/71.4	−61	585 ^g	0.22	1746	361
[HmIm][TFSI]	167/280	185.9/159.2	−6	68 ^h	2.7	566	336

^aCalculated using densities of constituents, $d_{\text{PSS}} = 1.44 \text{ g/cm}^3$, $d_{\text{PS}} = 1.05 \text{ g/cm}^3$, and $d_{\text{PMB}} = 0.86 \text{ g/cm}^3$, by ignoring the volume change of mixing.

^bAt 25 °C, measured using small-angle X-ray scattering in the absence of ionic liquids. ^cMeasured using Nano UTM. ^dvan der Waals volume; calculated using ChemAxon's Calculator Plugins. ^eMeasured using differential scanning calorimetry (DSC). ^fAt 20 °C, ref 48. ^gAt 25 °C, ref 49. ^hAt 25 °C, ref 50. ⁱAt 25 °C, measured in our lab. ^j*Ab initio* calculation at 0 K in a vacuum; calculations were performed using a DFT exchange–correlation functional, B3LYP, with 6-31G(d) basis set. ^k[HmIm][−SO₃H] = 383 kJ/mol.

appropriate ILs, as determined by the combination of cations and anions, is key to improve the actuation performance. Considerable attention has thus been paid to imidazolium-based ILs by focusing on chemical substituents at the 1- and 3-positions of the imidazole ring,^{19,20} accompanied by varying the counteranions.^{15,17,18}

From a number of systematic studies on EAP actuators composed of ionic polymers and imidazolium ILs, two main conclusions have been drawn. First, the type of cation and anion in the IL is the most important factor in determining the actuation performance by influencing the ion–ion interactions (binding energy), ionic conductivity, charging time, and degree of volume change under an electric field stimulus.²⁰ The type of IL also has an important bearing on the glass transition temperature (T_g) of IL-containing polymers by influencing the formation of ionic clusters.^{21,22} Second, a simple means of improving the ionic conductivity of IL-containing polymers, which is closely associated with the ion diffusion behavior and charging time of the actuators, is to increase the amount of IL in the polymers.²³ However, this leads to a significant deterioration of the mechanical properties, obstructing durable bending under prolonged exposure to voltage switching.^{24,25} Consequently, throughout the literature, the interplay between ionic conductivity and mechanical stability of IL-containing polymers has often been taken into account for achieving the optimized actuation performances.^{22,23}

Many previous studies on IL-containing polymers for actuator applications treated the matrix polymer merely as a support layer for ion transport.^{26,27} This led to the exclusive use of a few commercially available polymers in EAP actuators, i.e., poly(vinylidene fluoride-co-hexafluoropropene) (PVdF-HFP)^{15–17} and perfluorosulfonic acid ionomers (Nafion and Aquivon).^{28,29} From a short-term perspective, this approach is certainly advantageous to obtain valuable outcome; however, the development of new polymers will ultimately be required to advance the technology. From this standpoint, several groups have recently attempted to employ new ionic polymers in EAP actuators to attain synergy between the polymer matrix and the IL.^{30–32} Microphase-separated ionic polymers have in particular emerged as promising candidates, inspired by theoretical predictions of facilitated fast ion transport along well-organized ionic channels.^{33,34} In fact, this prediction has already been proven by a body of literature on IL-containing block copolymers, where improved ionic conductivity at mitigated reduction of mechanical strength was demonstrated.^{35–41} These properties are attributed to the confinement of ILs inside the nanoscale ionic domains, which appear to reduce the

ion–cluster formation,^{42,43} accompanied by the construction of mechanically robust nonionic phases positioned every few tens of nanometers.

Long et al.^{23,31,32,44} and Koo et al.^{45,46} reported uses of microphase-separated ionic polymers in EAP actuators. Long and co-workers improved the actuation properties (good mechanical integrity and fast actuation response) as well as achieved rational molecular designs of self-assembled ionic polymers. Examples of such polymers include sulfonated pentablock copolymers,³¹ imidazolium-tethered block copolymers,²³ and zwitterionomers with tunable ionic interactions.⁴⁴ Koo and co-workers also exploited phase-separated sulfonated pentablock copolymers in EAP actuators for enhancing the actuation performance. Intriguingly, the integration of ILs, in conjunction with surface-modified inorganic fillers, into nanoscale ionic domains eliminated back-relaxation by allowing both fast ion transport and good mechanical strength.⁴⁶ The aforementioned results imply that the potential for EAP actuators comprising ILs and self-assembled ionic polymers could be immense, but the understanding of the factors of ionic polymers relevant for improving the actuation properties of IL-containing polymers is in its infancy.

Here, we investigate a new type of EAP actuator based on nanostructured ionic polymers and imidazolium ILs. A set of linear diblock and triblock copolymers composed of sulfonated polystyrene (PSS) and poly(methylbutylene) (PMB) has been synthesized to elucidate how parameters such as block architecture, molecular weight, and Young's modulus affect actuation properties. Notably, we proved that the bending strain and durability of the actuators are tunable in a straightforward manner by controlling the block architecture and molecular weight. The extent of electromechanical deformation and charging time of the actuators further appeared to be largely affected by molecular interactions between the polymer matrix and ILs, as determined by the use of three different imidazolium ILs composed of dissimilar anions.

The motivation of the present study is to unveil the key factors affecting the electromechanical properties of EAP actuators composed of self-assembled ionic polymers and ILs. A combination of *in situ* scattering and *in situ* spectroscopy experiments under voltage stimuli elucidates a molecular-level understanding of the bending mechanisms of our actuators. Our work should establish a bridgehead to expand the classes of ionic polymers suitable for future electrochemical devices.

EXPERIMENTAL SECTION

Synthesis of PSS–PMB and PSS–PMB–PSS Block Copolymers. A set of di- and triblock copolymers composed of polystyrene (PS) and polyisoprene (PI) were synthesized by sequential anionic polymerization of styrene and isoprene using cyclohexane as a solvent and *sec*-butyllithium as an initiator. For PS–PI–PS triblock copolymer synthesis, the styrene monomer was again brought into the reaction mixture containing PS–PI after complete consumption of the isoprene monomer. The selective hydrogenation of PI chains of the copolymers was then performed in the presence of a homogeneous Ni–Al catalyst in cyclohexane, using a 2 L Parr batch reactor at 80 °C and 420 psi. The hydrogenated block copolymers are referred to as PS–PMB and PS–PMB–PS, where PMB denotes polymethylbutylene. The molecular weight and molecular weight distribution of PS–PMB and PS–PMB–PS copolymers were characterized by combining ^1H nuclear magnetic resonance (^1H NMR, Bruker AVB-300) spectroscopy and gel permeation chromatography (GPC, Waters Breeze 2 HPLC) with PS standards in tetrahydrofuran (THF) for calibration. The polydispersity indices of the copolymers were less than 1.06. The PS chains of the saturated copolymers were partially sulfonated using 1 g of polymer, 1.8 mL of acetic anhydride, 0.5 mL of 96% sulfuric acid, and 40 mL of 1,2-dichloroethane at 40 °C under a N_2 blanket.⁴⁷ Samples with sulfonation level (SL) of 42 ± 3 mol % were obtained with 3 h of sulfonation reaction, as determined by ^1H NMR spectra with acetone- d_6 . The characteristics of resulting materials are listed in Table 1.

Ionic Liquids (ILs). 1-Hexyl-3-methylimidazolium bis-(trifluoromethane)sulfonimide ([HMIm][TFSI], $\geq 98\%$), 1-hexyl-3-methylimidazolium hexafluorophosphate ([HMIm][PF₆], $\geq 98\%$), and 1-hexyl-3-methylimidazolium tetrafluoroborate ([HMIm][BF₄], $\geq 97\%$) were purchased from Sigma-Aldrich and used as received. The molecular structures of the [HMIm] cation, [TFSI] anion, [PF₆] anion, and [BF₄] anion are shown in Figures 1c, 1d, 1e, and 1f, respectively. The characteristics of each IL are listed in Table 1. In order to measure solvation time of each IL, time-resolved fluorescence

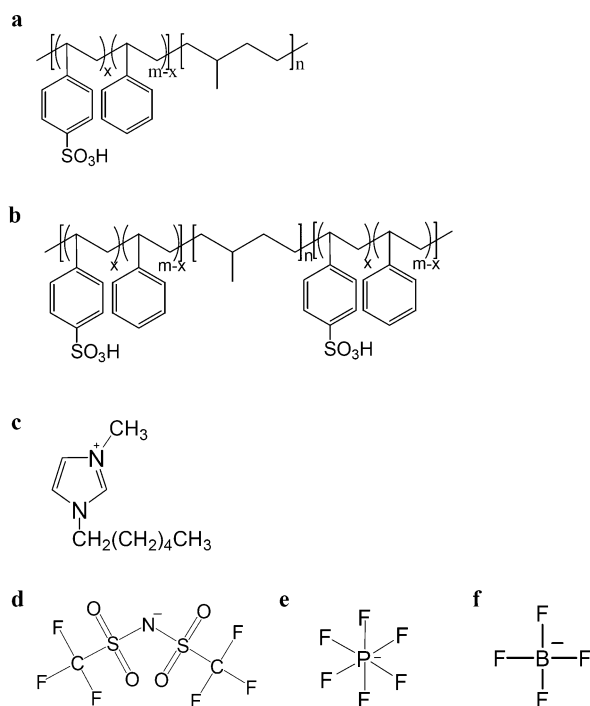


Figure 1. Molecular structures of (a) poly(styrenesulfonate-*b*-methylbutylene) (PSS–PMB), (b) poly(styrenesulfonate-*b*-methylbutylene-*b*-styrenesulfonate) (PSS–PMB–PSS), (c) 1-hexyl-3-methylimidazolium, [HMIm], (d) bis(trifluoromethane)sulfonimide, [TFSI], (e) hexafluorophosphate [PF₆], and (f) tetrafluoroborate [BF₄].

spectra of ILs were recorded by the time-correlated single photon counting (TCSPC) method. Laser grade coumarin 153 (C153) from Exciton was used without further purification, and the measurements were performed at 25 °C.

Preparation of IL-Containing Polymer Membranes. Inhibitor-free anhydrous THF ($\geq 99.9\%$) was used without further purification, and methanol was degassed 3 times prior to use. In an Ar-filled glovebox with oxygen and moisture concentration below 0.1 ppm, predetermined quantities of ILs and block copolymers were weighed into glass vials, and ca. 5 wt % solutions were prepared using 50/50 vol % THF and methanol mixture. The use of methanol allows the selective incorporation of ILs within PSS phases, which was confirmed by monitoring T_g values of PSS and PMB chains upon the addition of ILs. IL-containing polymer membranes with 100 μm thickness were prepared by solvent casting followed by vacuum drying at 70 °C for 7 days to remove any residual solvents.

Preparation of Actuators. Single-walled carbon nanotubes (SWCNTs, produced by a CoMoCAT catalytic CVD process) were purchased from Sigma-Aldrich and used without further purification. In order to prepare SWCNTs electrodes, 82 mg of ILs, 50 mg of PVdF-HFP (Kynar Flex 2801, Arkema Chemical Inc.), and 33 mg of SWCNTs were dissolved in 6 mL of *N,N*-dimethylacetamide (DMAc, $\geq 99.9\%$, Sigma-Aldrich). The mixture was stirred at room temperature for 24 h and placed in an ultrasonic bath for 6 h. The electrode layer was fabricated by casting the mixture onto a Teflon plate (2 cm $\times$ 2 cm) at room temperature, followed by vacuum drying at 80 °C. The thickness of the obtained electrode film was 20–30 μm . The polymer membrane was sandwiched between two SWCNTs electrodes via hot pressing. Rectangular pieces (8 mm $\times$ 1 mm) were cut from the hot-pressed sample where the final structure of the actuators was examined by scanning electron microscope (SEM, a Helios Nanolab 600) with a 5 keV accelerating voltage.

Actuator Performance Tests. The 8 mm $\times$ 1 mm $\times$ 120 μm sized actuator strips were clipped by two Pt disk electrodes yielding 6 mm free length from Pt contacts. Actuator performance was tested by applying square wave voltages to the actuator strips using a function generator (ED Corp., FG-1880B) and an oscilloscope (Agilent Technologies Inc., DSP3202A). The time-dependent bending motion of the actuators under voltage stimuli was recorded using a Zeiss microscope (AxioVision) equipped with a CCD camera. All measurements were carried out under ambient lab atmosphere with average relative humidity, RH = $30 \pm 5\%$.

Small-Angle X-ray Scattering (SAXS). The IL-containing polymers were laminated into an airtight sample cell, composed of an aluminum spacer, two Kapton windows, O-rings, and aluminum covers. Synchrotron SAXS measurements on these samples were performed using the PLS-II 4C SAXS beamline at the Pohang Accelerator Laboratory (PAL). The wavelength (λ) of the incident X-ray beam was 0.15 nm ($\Delta\lambda/\lambda = 10^{-4}$), and a sample-to-detector distance of 2.0 m was used, yielding scattering wave vector q ($q = 4\pi \sin(\theta/2)/\lambda$, where θ is the scattering angle) in the range 0.03–2.4 nm⁻¹. The resulting two-dimensional scattering data were averaged azimuthally to obtain intensity versus q .

In Situ SAXS. *In situ* SAXS measurements on the actuators were performed using the PLS-II 9A U-SAXS beamline at the PAL equipped with the function generator and the oscilloscope. The X-ray beam size at the sample position was as small as 30 μm , and the sample was mounted on the motorized 7-axis sample stage for the fine alignment in ± 1.0 μm step. This is to examine different sample areas beneath the electrodes during actuation. Considering X-ray beam size, we fabricated the actuators composed of relatively thick polymer layers (ca. 150 μm thick) for accurate scanning of three independent areas. The wavelength (λ) of the incident X-ray beam was 0.1119 nm ($\Delta\lambda/\lambda = 2 \times 10^{-4}$), and a sample-to-detector distance of 4.5 m was used, yielding q in the range 0.04–1.03 nm⁻¹. The resulting scattering data were collected using 2D CCD detector (Rayonix SX16S, USA) and averaged azimuthally to obtain scattering intensity versus q .

Transmission Electron Microscopy (TEM). The IL-containing polymer membranes were cryo-microtomed at –120 °C to obtain thin sections with thicknesses in the 80–120 nm range using a RMC

Boeckeler PT XL ultramicrotome. The electron contrast in the samples was enhanced by exposure to ruthenium tetroxide (RuO_4) vapor for 50 min. TEM images of stained samples were obtained with a Hitach H-800 microscope operating at 100 kV. All data sets were acquired using Digital Micrograph (Gatan, Inc.) software.

Conductivity Measurements. The conductivities of IL-containing polymers were measured by combining a potentiostat (PARSTAT 2273) and an ac impedance spectroscopy (Solartron 1260A) in a glovebox. The through-plane conductivity was measured using a home-built two-electrode cell with $1.25\text{ cm} \times 1.25\text{ cm}$ stainless steel blocking electrodes, Kapton spacers, and $1\text{ cm} \times 1\text{ cm}$ Pt working and counter electrodes. The counter electrode was engraved with $0.8\text{ cm} \times 0.8\text{ cm}$ hole to load the samples ($0.8\text{ cm} \times 0.8\text{ cm} \times 200\text{ }\mu\text{m}$). Data were collected over a frequency range of 1–100000 Hz.

Capacitance Measurements. A dc step voltage of 3 V was applied to the IL-containing polymer membranes ($3\text{ mm} \times 3\text{ mm} \times 50\text{ }\mu\text{m}$) for 40 s using a potentiostat (VersaSTAT3, Princeton Applied Research, AMETEK, Inc.), and then the voltage was set to zero. The time-dependent charging current was recorded and integrated to obtain the stored charge in the samples. Upon dividing the charge by interfacial area of the sample, the unit area charge [C/cm^2] was estimated.

Nano Universal Testing Machine (Nano UTM). The tensile test was performed using a Nano UTM (MTS Nano Instruments) to measure Young's moduli of the IL-containing polymer membranes. The IL-containing polymer membranes with average thickness of $60\text{ }\mu\text{m}$ were cut into thin rods (15 mm long and $200\text{ }\mu\text{m}$ wide). The membrane was tied up by paper and was left to stand for more than 6 h to be fastened the rod together with paper. The specimen was then mounted and gripped on bolt jigs. Stress–strain measurements were carried out with 0.00027 strain per second of cross-head speed until a fracture of the specimen occurred (with a maximum load of 500 mN and a resolution of 50 nN). All measurements were carried out under ambient lab atmosphere with average relative humidity of $\text{RH} = 30 \pm 5\%$.

In Situ Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy (In Situ ATR-FTIR). ATR-FTIR spectra were measured using a Spectrum Two IR spectrometer (PerkinElmer), equipped with a HeNe laser. $100\text{ }\mu\text{m}$ thick IL-containing polymer membrane was sandwiched between two gold leaves (L.A. Gold Leaf, 50 nm) via hot pressing at $80\text{ }^\circ\text{C}$ for 5 min. A strip of the trilaminar membrane with dimension of $1.5\text{ mm} \times 7\text{ mm}$ was clamped onto a diamond crystal of the ATR-FTIR apparatus equipped with a pressure arm (L1600107, PerkinElmer) to ensure good contact of the sample with the ATR crystal. Dc voltages of $\pm 3\text{ V}$ were applied to the samples during the measurements to monitor ion migration near the gold/polymer interfaces.

Self-Diffusion Coefficient Measurements. The IL-containing polymer membranes were loaded into 4 mm (o.d.) NMR microtubes in an Ar-filled glovebox and were sealed with caps. Self-diffusion coefficients of the samples were measured using ^1H and ^{19}F pulsed gradient spin echo (PGSE) NMR (Bruker AVB-300) in a 7 and 1.5 T superconducting magnet, respectively, with z-axis gradient. Time interval of the field gradient and duration time between two gradient pulses were in the range 1–10 ms and 0.1–0.3 s, respectively. The sample temperature was calibrated using ethylene glycol standard. At each temperature, the samples were equilibrated for 1 h before the measurements.

RESULTS

Sulfonated Block Copolymers and ILs Used in the Present Study. The present study is based on three different linear AB diblock and ABA triblock copolymers composed of sulfonated polystyrene (PSS) and poly(methylbutylene) (PMB). The molecular structures of the diblock and triblock copolymers are shown in Figures 1a and 1b, respectively. The characteristics of the polymers are summarized in Table 1. The polymers are referred to as D17(42), T7(45), and T12(40), where D and T signify diblock and triblock architectures,

respectively. The integers 17, 7, and 12 indicate the nominal molecular weights of PS precursors in kg/mol. The sulfonation level (SL) of each sample is given in parentheses, which was fixed at $42 \pm 3\text{ mol } \%$ by controlling the sulfonation reaction time. This is to elucidate the effects of block architecture and polymer molecular weight on actuation properties of IL-containing polymers while excluding the effects of local concentrations of ions. All the three copolymers exhibited analogous hexagonally packed cylindrical (HEX) morphologies in the absence of ILs. Note that the domain size of D17(42) was markedly larger than those of T7(45) and T12(40) (Table 1), presumably owing to the loop and/or bridge formation of PMB middle blocks for T7(45) and T12(40) copolymers when they are packed into ordered grains.

Covalently bonded nonionic PMB chains are expected to retain the mechanical integrity of the polymer upon the addition of ILs. We found that the Young's moduli of sulfonated block copolymers were largely affected by the block architecture, as listed in Table 1; the modulus of T7(45) was as high as 3.2 GPa, compared to the relatively low value of 2.1 GPa for D17(42). The high-molecular-weight T12(40) exhibited the largest modulus of 4.2 GPa. Thus, a triblock architecture is advantageous in improving the mechanical integrity of polymers.

Figures 1c–f show the chemical structures of cations and anions in the ILs used in the present study. Three different anions—bis(trifluoromethane)sulfonimide [TFSI], hexafluorophosphate [PF_6], and tetrafluoroborate [BF_4 —were employed while maintaining the same 1-hexyl-3-methylimidazolium [HMIIm] cation. The characteristics of the three ILs, [HMIIm]-[TFSI], [HMIIm][PF_6], and [HMIIm][BF_4], are also summarized in Table 1. Our study has revealed that the type of anion in the IL impacts the melting temperature, viscosity, binding energy, and ionic conductivity of the IL. The large [TFSI] anion enhances the ionic conductivity of the IL, while the use of [PF_6] results in an order of magnitude lower conductivity. It should be noted that because the [TFSI] anion is weakly bound to the [HMIIm] cation compared to [$-\text{SO}_3^-$] in the polymer, the sulfonate groups localized within the PSS phases of the HEX structure could promote an effective dissociation of [HMIIm][TFSI]. Conversely, a higher binding energy was found for [BF_4] and [HMIIm] compared with that of [$-\text{SO}_3^-$] and [HMIIm], which denotes a marginal ability to expedite molecular interactions between the IL and the polymer matrix.

Effects of Anion Type in ILs on Actuation Performance. We first explored the effects IL type on the actuation properties of actuators based on IL-containing sulfonated block copolymers. To begin, the morphologies of IL-containing copolymers by varying the anion in the IL were examined by small-angle X-ray scattering (SAXS). Figure 2 shows a set of scattering profiles for D17(42) copolymers incorporated with 30 wt % IL, compared with that of neat D17(42), measured at $25\text{ }^\circ\text{C}$. By incorporating each of the three ILs in D17(42), we observed a series of Bragg peaks at $1q^*:\sqrt{3}q^*:\sqrt{4}q^*:\sqrt{7}q^*:\sqrt{9}q^*:\sqrt{11}q^*:\sqrt{12}q^*$ with $q^* = 2\pi/d_{100}$, indicating the formation of analogous HEX morphologies. Note that while we marked all allowed reflections of the HEX structure in Figure 2, some of the reflections were poorly resolved, particularly for the [HMIIm][TFSI]-containing sample. It is also noted that the form factor minimum reduces the intensities of the $\sqrt{7}$ peak of neat D17(42), as denoted by gray color.

Interestingly, the domain size (d_{100}) of the IL-containing D17(42) copolymers was found to be sensitive to the type of

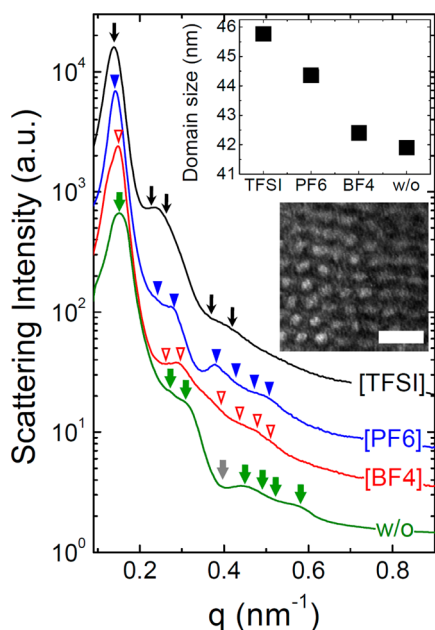


Figure 2. SAXS profiles of D17(42) membranes with the addition of 30 wt % [HMIIm][TFSI], [HMIIm][PF₆], and [HMIIm][BF₄], compared with that of neat D17(42), measured at 25 °C. The scattering profiles are vertically offset for clarity. The regular inverted arrows, the inverted filled triangles, the inverted open triangles, and the inverted thick arrows represent Bragg peaks at the q ratios of $1:\sqrt{3}:\sqrt{4}:\sqrt{7}:\sqrt{9}:\sqrt{11}:\sqrt{12}$, indicative of HEX morphology. The domain size of each sample is plotted in the inset. A cross-sectional TEM image of [HMIIm][BF₄]-containing D17(42) given in the inset confirms PMB cylinders dispersed in PSS phases (darkened by RuO₄ staining). The scale bar represents 100 nm.

anion in the IL, as shown in the inset of Figure 2. The domain size increases in the order [HMIIm][TFSI] > [HMIIm][PF₆] > [HMIIm][BF₄], which corresponds to the order of increase in the van der Waals volume of the anion in the IL (Table 1), implying that tunable swelling of the PSS phase by adjusting the type of IL. Note that although we only discuss the representative data obtained with the D17(42) copolymer, analogous conclusions were found for T7(45) and T12(40) copolymers (data not shown here). Additionally, qualitatively similar Bragg peaks indicative of HEX morphology were observed for all IL-containing copolymers with different IL concentrations in the range 0–70 wt % where larger amounts of IL resulted in larger domain sizes.

The HEX morphology of the IL-containing sulfonated block copolymers was confirmed by cross-sectional TEM on the cryo-microtomed membrane. The representative TEM image obtained with [HMIIm][BF₄] is given in the inset of Figure 2, showing the HEX structure possessing nonionic PMB cylinders dispersed in the IL-filled PSS phases (the PSS phases were darkened by RuO₄ staining). The domain size obtained from the TEM images was in good agreement with the SAXS data. It should be noted that we intentionally adopted a HEX structure with a major PSS phase, which is advantageous for achieving fast ion migration under a given electric field as the ions can effectively diffuse along the continuous ionic domains without the need for aligning the PSS phase.³

We fabricated actuators consisting of IL-containing D17(42) copolymers sandwiched between SWCNT electrodes. The dimension of the actuator strips was 2 mm × 8 mm × 120 μm. A typical cross-sectional SEM image of the actuators with

monolithic trilaminar structures is shown in Figure 3a. Figure 3b shows the time-dependent bending response of the

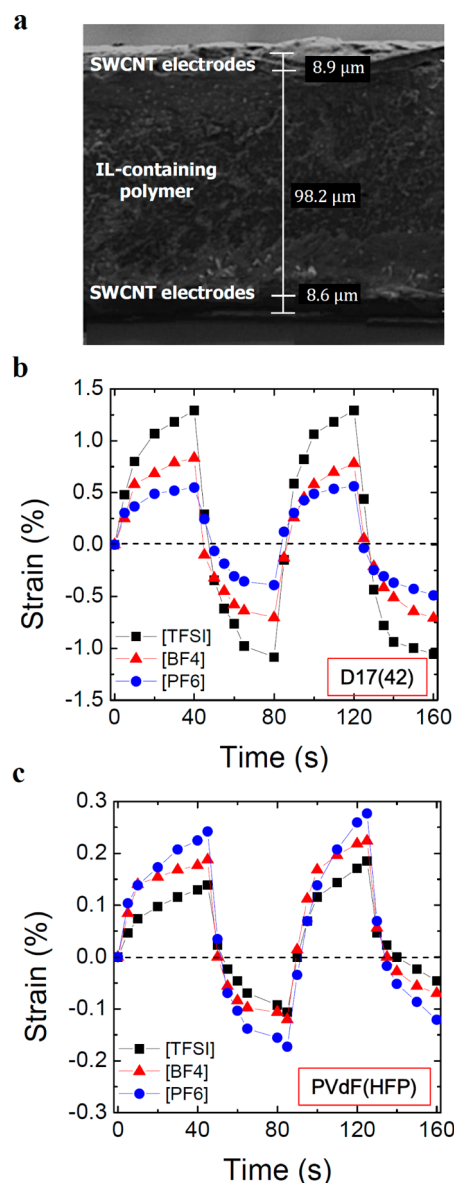


Figure 3. (a) Cross-sectional SEM micrograph of the actuator composed of IL-containing polymer layer and SWCNT electrodes. The bending strain of actuators based on (b) IL-containing D17(42) and (c) IL-containing PVdF(HFP) at alternating square-wave voltages of ± 3 V and a frequency of 0.0125 Hz, by varying anions in the IL, measured at room temperature. For (b) and (c) a fixed IL concentration of 30 wt % was used.

actuators at alternating square-wave voltages of ± 3 V with a cycle time of 80 s, measured at room temperature and ambient lab atmosphere with average relative humidity, RH = $30 \pm 5\%$. The strain was calculated at measured displacements from the position of the actuator tip, where a negative strain indicates bending of the actuator in the opposite direction under -3 V. As can be seen from Figure 3b, when the anion in the IL was varied from [TFSI] to [PF₆] while keeping the same [HMIIm] cation, more than a 2-fold reduction in strain was observed for the same IL loadings (30 wt %). Such behavior was consistently observed for actuators composed of IL-containing T7(45) and IL-integrated T12(40) copolymers.

Our observations are intriguing because, to the best of our knowledge, there are no reports on the better actuation performance using [TFSI]-based ILs compared to [PF₆]- and [BF₄]-containing ILs.^{15,17,49} The known shortcomings is the bulkiness of the [TFSI] anion; the difference in van der Waals volume between [HMIIm] and [TFSI] is only 26.7 Å³ (Table 1), which hindered the generation of high strain as a result of ion migration. Note in passing that the actuation properties acquired under Ar-filled glovebox were analogous to the results given in Figure 3b (data not shown here), signaling that the hygroscopic nature of the samples had little effect on the conclusion.

To confirm the IL-dependent electromechanical deformation of IL-containing sulfonated block copolymers, the time-dependent bending response of actuators based on conventional PVdF(HFP) was measured under the same conditions with the addition of the same amount of IL (30 wt %). As shown in Figure 3c, for the PVdF(HFP) copolymer, the bending behavior was markedly altered with the largest strain now seen with [HMIIm][PF₆]. Notably, in contrast to IL-containing sulfonated block copolymers, the smallest bending strain was found for the [HMIIm][TFSI]-containing PVdF(HFP) copolymer, consistent with the literature.^{15,17,51}

More remarkably, the actuators composed of IL-containing D17(42) copolymers demonstrated a 2–10-fold increase in the bending strain compared to the actuators comprising PVdF(HFP) and the same type of IL (compare Figures 3b and 3c). The increase in the strain clearly indicates the importance of the polymer matrix in determining the actuation performance, where polymers possessing well-defined ionic phases seem to be more advantageous for improving the bending response by providing the organized ion migration along the less tortuous ion-conduction pathways. In addition, the confinement of ILs within the nanoscale PSS domains could reduce the formation of ionic clusters^{43,52} with an aid of highly interactive –SO₃H surfaces, thereby increasing the ion transport efficiency.

To understand the unusual IL anion-dependent actuation performance of the actuators made with the IL-containing sulfonated block copolymers, we carried out conductivity, capacity, and modulus measurements for the IL-containing polymers. We first examined the effects of IL type on the ionic conductivity of IL-containing sulfonated block copolymers. Figure 4a shows ionic conductivities of IL-containing D17(42) at an IL concentration of 30 wt % with varying anions in the IL. While the temperature-dependent conductivities are plotted, the practically important one should be room-temperature conductivity given that the actuator is operated at ambient conditions.

As clearly revealed by the Figure 4a, the highest room-temperature conductivity was obtained with [HMIIm][TFSI], whereas that with [HMIIm][PF₆] was 2 orders of magnitude lower. This is likely associated with the intrinsic conductivity of neat ILs (Table 1); however, such a large reduction in conductivity (on the order of 10^{−5} mS/cm at 25 °C) for [HMIIm][PF₆]-embedded polymers is rare. This assessment stems from the fact that the conductivity of [HMIIm][PF₆]-containing PVdF(HFP) is on the order of 10^{−2} mS/cm at 25 °C for the same IL concentration. It is thus inferred that the molecular interactions between the IL and the polymer matrix significantly altered the diffusion behavior of the ILs. The high conductivity obtained with [HMIIm][BF₄] is attributed to the low *T_g* of the membrane (40 °C lower than that of [HMIIm][TFSI]-containing membrane). Note that the activa-

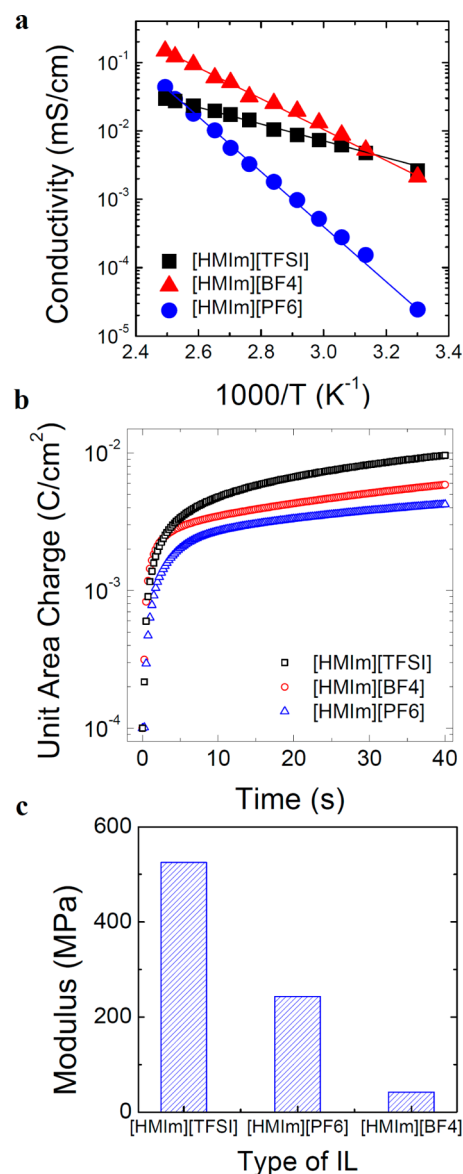


Figure 4. (a) Temperature-dependent ionic conductivities, (b) time-dependent charge density with 3 V step voltage, and (c) Young's moduli of D17(42) membranes upon incorporating [HMIIm][TFSI], [HMIIm][BF₄], and [HMIIm][PF₆] at an IL concentration of 30 wt %. The linear fits in (a) were obtained by Arrhenius analysis.

tion energy (*E_a*) of IL-containing D17(42) obtained by Arrhenius analysis differed greatly (23.5, 43.3, and 80.6 kJ/mol for [HMIIm][TFSI], [HMIIm][BF₄], and [HMIIm][PF₆], respectively), contrary to the similar *E_a* values of IL-containing PVdF(HFP) (31.4 for [HMIIm][TFSI], 34.1 for [HMIIm][BF₄], and 36.7 kJ/mol for [HMIIm][PF₆]).

We further investigated the charging behavior of each actuator to elucidate the time-dependent ion migration in the direction of oppositely charged electrode surfaces under an applied voltage. A dc step voltage of 3 V was applied to IL-containing polymers for 40 s. Figure 4b shows the unit area charge as a function of time for the IL-containing D17(42) copolymers, where the highest unit area charges were acquired when employing [HMIIm][TFSI], followed by [HMIIm][BF₄] and [HMIIm][PF₆], which is in good agreement with the bending strain of the actuators (Figure 3b). The slow ion migration at the beginning of the charging cycle for the

[HMIIm][PF₆]-containing copolymer is also consistent with conductivity results (Figure 4a).

We now discuss the effects of IL type on the mechanical properties of IL-containing sulfonated block copolymers. For accurate measurements of the stress–strain behavior of actuator strips with only a few millimeter dimensions, a Nano UTM was employed.⁵³ The resulting Young's moduli of IL-containing D17(42) copolymers at 30 wt % IL loadings are plotted in Figure 4c. Overall, the addition of IL into D17(42) reduced the storage moduli, though the amount was largely governed by the type of anion in the ILs; the storage moduli were 530, 240, and 40 MPa for [HMIIm][TFSI], [HMIIm][PF₆], and [HMIIm][BF₄], respectively. It is noteworthy that the trend in the modulus of IL-containing D17(42) copolymers is not consistent with that of the viscosity of neat ILs, [HMIIm][PF₆] = 585 cP > [HMIIm][BF₄] = 314 cP > [HMIIm][TFSI] = 68 cP, as listed in Table 1. The achievement of high mechanical strength with low-viscosity [HMIIm][TFSI] is thus notable, enabling us to attain both high ionic conductivity and good mechanical properties of IL-containing polymer layers without considering their interplay.

Effects of Block Architectures and Molecular Weights on Actuation Performance. Next, we elucidate the factors affecting the actuation performance of IL-containing sulfonated block copolymers by focusing on polymer-related parameters. The type of IL was fixed to [HMIIm][TFSI] to guarantee the best actuation performance. Figure 5 shows the representative SAXS profiles of [HMIIm][TFSI]-containing copolymers at 60

wt % IL loadings with varying polymers of D17(42), T7(45), and T12(40). Analogous HEX morphologies with qualitatively similar Bragg peaks ($\downarrow$) at the q ratios of $1:\sqrt{3}:\sqrt{4}:\sqrt{7}:\sqrt{9}:\sqrt{11}:\sqrt{12}$ were revealed after the incorporation of [HMIIm][TFSI] into each polymer. However, largely dissimilar domain sizes, depending on the block architecture and molecular weight, were found, as plotted in the inset of Figure 5. The high-molecular-weight characteristics of T12(40) are likely responsible for the relatively poor long-range ordering of the [HMIIm][TFSI]-containing T12(40) copolymer. Representative TEM images obtained with [HMIIm][TFSI]-containing T7(45) and T12(40) are also shown in the insets to confirm the HEX morphology, where the PSS phases were darkened by RuO₄ staining.

Figure 6a shows the peak-to-peak bending strains of the actuators based on D17(42), T7(45), and T12(40) copolymers

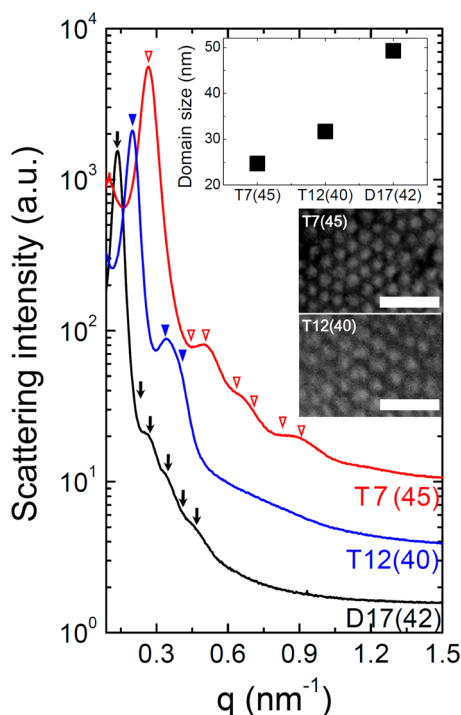


Figure 5. SAXS profiles of D17(42), T12(40), and T7(45) with the addition of 60 wt % [HMIIm][TFSI], measured at 25 °C. The scattering profiles are vertically offset for clarity. The arrows ($\downarrow$), the inverted filled triangles (∇), and the open triangles ($\triangledown$) represent Bragg peaks at q ratios of $1:\sqrt{3}:\sqrt{4}:\sqrt{7}:\sqrt{9}:\sqrt{11}:\sqrt{12}$, indicating the HEX morphology. The domain size of each sample and cross-sectional TEM images obtained with [HMIIm][TFSI] and T7(45) (and T12(40)) are given in the insets. The scale bars represent 100 nm.

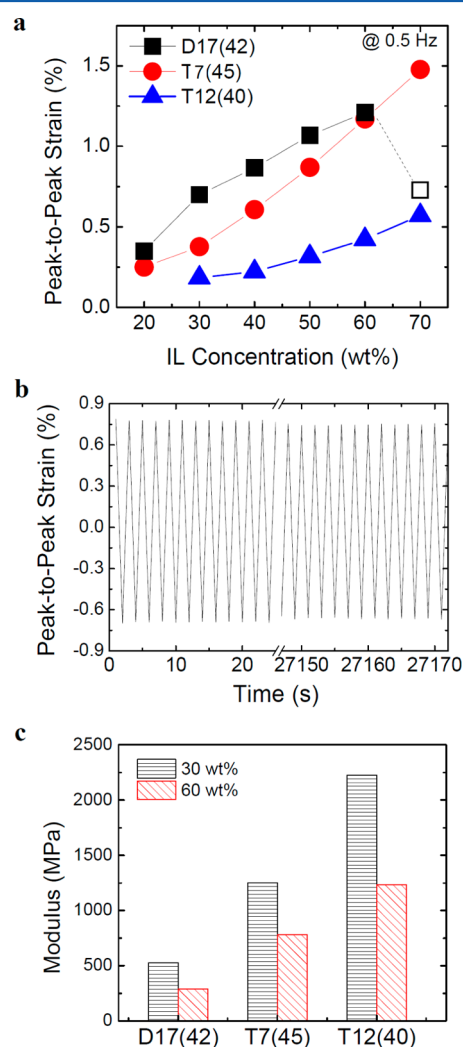


Figure 6. (a) Peak-to-peak bending strains of actuators based on D17(42), T7(45), and T12(40) containing [HMIIm][TFSI], measured at a cycle time of 2 s, by varying the IL loading. The open black square represents the average strain value of 70 wt % IL-containing D17(42) with 10 cycles (b) Cycle life of the actuator strip made of T7(45) copolymer integrated with 70 wt % [HMIIm][TFSI] under continuous operation in air at alternating square-wave voltages of ± 3.0 V with a frequency of 0.5 Hz. (c) Young's moduli of D17(42), T7(45), and T12(40) copolymers containing [HMIIm][TFSI]. The amount of IL loadings is noted in the figure.

integrated with [HMIIm][TFSI] at alternating square-wave voltages of ± 3 V and 0.5 Hz, measured at room temperature. At low IL concentrations (below 50 wt %), better actuation performances were obtained with diblock architecture than with triblock. However, at high IL loadings (above 60 wt %), the bending strain of the D17(42) diblock copolymer was reduced by repeated bending. This is contrary to the durable bending found for actuators composed of T7(45) and T12(40) triblock copolymers and 70 wt % [HMIIm][TFSI]. Thus, we conclude that the use of sulfonated block copolymers with triblock architecture is beneficial in improving the actuator performance by enabling the addition of large amounts of ILs with retained mechanical integrity. As a result, the T7(45) triblock copolymer demonstrated the highest peak-to-peak strain of 1.5% with 70 wt % IL loadings. However, the degree of electromechanical deformation of IL-containing T12(40) was markedly low compared to the actuator based on the T7(45) triblock copolymer, indicating the impact of molecular weight of the triblock copolymer on the actuation response. A greater amount of IL loading into both the T7(45) and T12(40) copolymers may further improve the actuation performance; however, we did not devote efforts to confirm this.

Figure 6b demonstrates the long-term durability of the actuators made of 70 wt % [HMIIm][TFSI]-containing T7(45) triblock copolymer. We measured the bending strain in response to ± 3.0 V square wave input signals at a frequency of 0.5 Hz. By switching the electric field every 1 s, it was revealed that actuation can be repeated over 13 000 cycles with negligible changes in actuator stroke or restoration time at a negative potential. This implies reversible and fast ion diffusion within the T7(45) matrix when an electric field is applied. This is in sharp contrast to many ionic polymer actuators reported in the literature, which show deterioration in the bending strain with repeated voltage switching, especially at high frequencies (>0.5 Hz). The poor durability primarily originates from ill-organized ion migration behavior because most of the matrix polymers employed to date possess discrete ionic domains rather than continuous phases.

The block architecture-dependent actuation performance was further understood by measuring the Young's moduli of IL-containing polymers. Figure 6c summarizes the moduli of D17(42), T6(45), and T12(40) copolymers obtained with 30 and 60 wt % [HMIIm][TFSI] loadings. It is evident that the mechanical strength of IL-containing polymers can be tuned by controlling the amount of IL and/or varying the block architecture/molecular weight. T12(40) demonstrated high mechanical strength (over 1 GPa) even with the addition of 60 wt % IL and the modulus of 60 wt % [HMIIm][TFSI]-containing T7(45) was also high at 780 MPa, contrary to the low value of 250 MPa found with D17(42) with the same IL loading. This is presumably a result of the formation of physical cross-links in the PMB domains of the triblock copolymers by bridge formation, which is essentially not affected by the incorporation of ILs. While the 250 MPa modulus is suitable for an electromechanical transducer (>100 MPa),²⁷ the value was rapidly reduced to an immeasurably low value at an IL concentration of 70 wt %. This should contribute to the failure in the bending motion of D17(42) with 70 wt % [HMIIm][TFSI], as discussed in Figure 6a.

DISCUSSION

Mechanisms Underlying the Improved Actuation Properties of Actuators Composed of IL-Containing

Sulfonated Block Copolymers. From the results obtained thus far, there is no doubt about the advantage of employing [TFSI]-based ILs and self-assembled sulfonated block copolymers with optimized block architecture and molecular weight to improve actuation properties, yet molecular-level understanding of the actuation mechanisms has not been elucidated. We thus carried out in-depth studies on the actuation mechanisms by combining pulsed field gradient spin-echo nuclear magnetic resonance (PGSE-NMR), *in situ* attenuated total reflectance Fourier transform infrared (*in situ* ATR-FTIR) spectroscopy, and *in situ* small-angle X-ray scattering (*in situ* SAXS) experiments.

PGSE-NMR experiments were first performed to study the ion diffusion behavior of IL-containing polymers. The spin-echo signal attenuation was acquired with two gradient pulses and self-diffusion coefficients of the cation (D^+) and the anion (D^-) in IL-containing polymers were independently measured at varying temperatures using ^1H and ^1F probes.⁵² As plotted in Figure 7a, for IL-containing T7(45) copolymers, an order of magnitude higher D^+ and D^- values were determined with [HMIIm][TFSI] than with [HMIIm][PF₆]. This is consistent with the large bending strain and fast bending response seen with [HMIIm][TFSI] compared to those obtained with [HMIIm][PF₆] (see Figure 3b). It has been also noted that

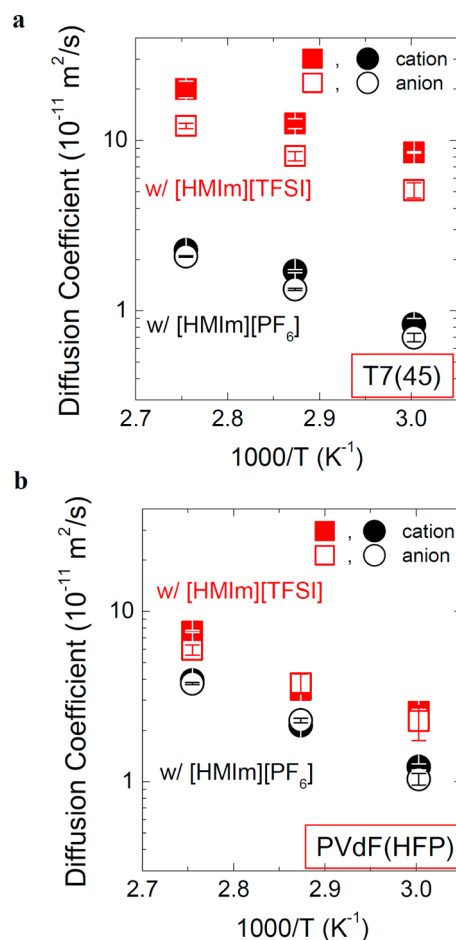


Figure 7. D^+ (filled symbols) and D^- (open symbols) values at different temperatures for (a) IL-containing T7(45) copolymers and (b) IL-containing PVdF(HFP) at an IL concentration of 30 wt %. The IL type was varied as [HMIIm][TFSI] (squares) and [HMIIm][PF₆] (circles), as indicated in the figures.

[HMIIm] diffusion is faster than [TFSI] diffusion while [HMIIm] and [PF₆] diffuse at relatively similar rates. Such differences were quantified by analyzing the cationic transference number, $T^+ = D^+/(D^+ + D^-)$, where 0.62 and 0.54 were obtained for [HMIIm][TFSI] and [HMIIm][PF₆], respectively. This indicates the prevalence of [TFSI] aggregates in the low-dielectric PSS phases, which reduces the Coulombic interactions between the [HMIIm] cation and the [TFSI] anion, which in turn enhances the ionic diffusion coefficient.⁵⁴

When the same experimental protocols were applied to IL-containing PVdF(HFP) as a control, fascinating differences in the diffusion behavior were obtained. As shown in Figure 7b, 3-fold reductions in the [HMIIm] and [TFSI] diffusion coefficients were determined for [HMIIm][TFSI]-embedded PVdF(HFP) compared to that of [HMIIm][TFSI]-containing T7(45). In contrast, the D^+ and D^- values of [HMIIm][PF₆]-embedded PVdF(HFP) increased, compared to those of [HMIIm][PF₆]-containing T7(45). The fact that cations and anions diffuse at similar rates for both ILs in PVdF(HFP) matrix is also noteworthy, as quantified by similar T^+ values of 0.53 and 0.54 for [HMIIm][TFSI] and [HMIIm][PF₆], respectively.

These results inspired us to further investigate the cation and anion migration behavior of IL-containing polymers under voltage stimuli using *in situ* ATR-FTIR experiments. The IL-containing polymers were sandwiched between 50 nm thick gold leaves so that a beam of IR light could be transmitted through the electrode/polymer interface. The IL concentration was fixed at 60 wt %. Figure 8a shows the time-resolved IR spectra at the interface of the gold leaf and [HMIIm][TFSI]-containing T7(45) in response to voltages of ± 3 V (+3 V creates a negatively charged electrode at the ATR crystal surface). The aromatic C–H stretching absorption band (3200–3050 cm⁻¹) from the [HMIIm] cation, along with the SO₂ (1350 cm⁻¹) and the CF₃ (1180 cm⁻¹) stretching vibrations from the [TFSI] anion, is assigned in the figure.⁵⁵ Note that the C=C (1570 cm⁻¹) and SO₃⁻ (1050 cm⁻¹) stretching vibrations of the [HMIIm] cation and [TFSI] anion, respectively, were excluded from the analysis because of the overlaps with those of the PSS chains. As clearly seen in the figure, at +3 V, the absorption intensities of both the [HMIIm] cation and [TFSI] anion increased with time. This indicates that, while the [HMIIm] cation is the charge carrier for actuation, mass transport occurred for both the [HMIIm] cation and [TFSI] anion. Our result is consistent with the recent work of Elabd et al. using an IL-embedded Nafion system.⁵⁶

Upon switching the direction of the voltage to -3 V, the IR spectra were restored to the initial profile (data not shown here), implying well-controlled movement of the [HMIIm] cation and [TFSI] anion under a given electric field. Considering that our actuators demonstrated an absence of back relaxation by displaying equilibrium displacement for up to several hours, we infer that ion migration and accumulation occurred in the actuators without reverse migration, as demonstrated by the stable absorbance intensities with over time for both the cation and anion. The large bending strains of our actuators made of [HMIIm][TFSI]-containing copolymers, despite the small van der Waals volume differences between [HMIIm] and [TFSI], can be understood by the accumulation of both cations and anions, dominated by cation diffusion toward oppositely the charged electrode surface.

We repeated the same experiments for [HMIIm][TFSI]-containing PVdF(HFP), as shown in Figure 8b. Again, both the

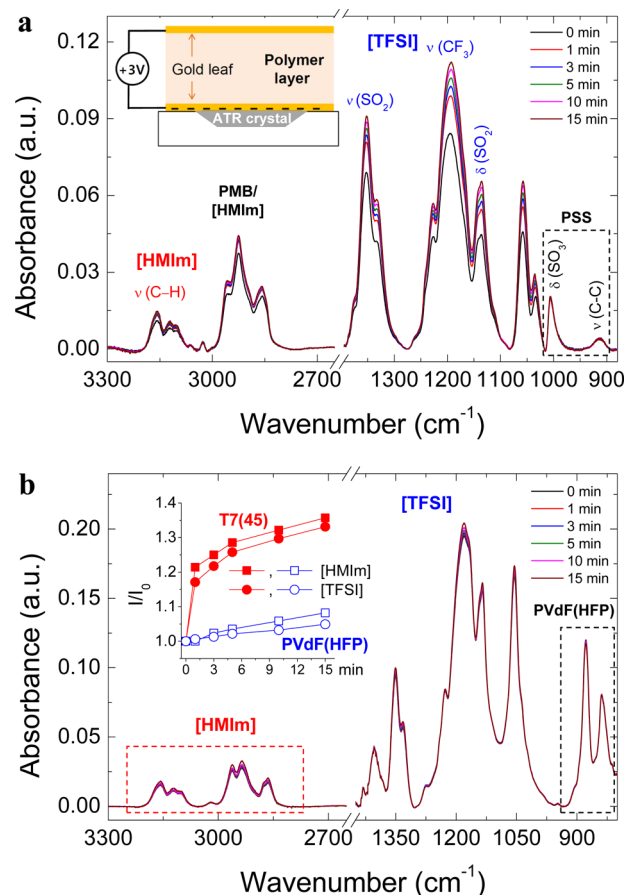


Figure 8. Time-resolved infrared spectra of [HMIIm][TFSI]-containing (a) T7(45) and (b) PVdF(HFP) in response to a voltage input of 3 V. Stretching vibrations for the [HMIIm] cation and [TFSI] anion are highlighted in red and blue colors, respectively, while those of polymers are given by black color. Schematic drawing of the experimental setup is shown in the inset of (a). The absorbance changes of the [HMIIm] cation and [TFSI] anion with time, relative to the initial values (I_0), of each sample are plotted in the inset of (b).

[HMIIm] cation and [TFSI] anion absorbance appeared to increase under +3 V, which decreased upon reversing the direction of electric field. However, the relative changes in the IR peak intensities of the [HMIIm] cation and [TFSI] anion were significantly smaller than those seen with T7(45) over the same time interval in response to voltage stimuli. This was quantified by plotting the intensity ratios I^{3158}/I_0^{3158} for the [HMIIm] cation and I^{1180}/I_0^{1180} for the [TFSI] anion, of each sample, as shown in the inset of Figure 8b. It was apparent that the degree of change in absorbance was markedly larger with T7(45) than with PVdF(HFP). The fast response (at 1 min) observed with [HMIIm]-containing T7(45) is also noteworthy and is responsible for the improved actuation performance of our actuators. The slow growth in IR peak intensities for [HMIIm][TFSI]-containing PVdF(HFP) signals the formation of ion clusters for ILs dispersed within the PVdF(HFP) matrix, which is consistent with neat ILs where the strong Coulombic interactions between the IL cation and the anion impede separation.⁵⁷ The difference between T7(45) and PVdF(HFP) stems from the existence of interactive $-\text{SO}_3^-$ surfaces on T7(45) offering a strong binding affinity with the [HMIIm] cation, thereby promoting effective dissociation of the IL.

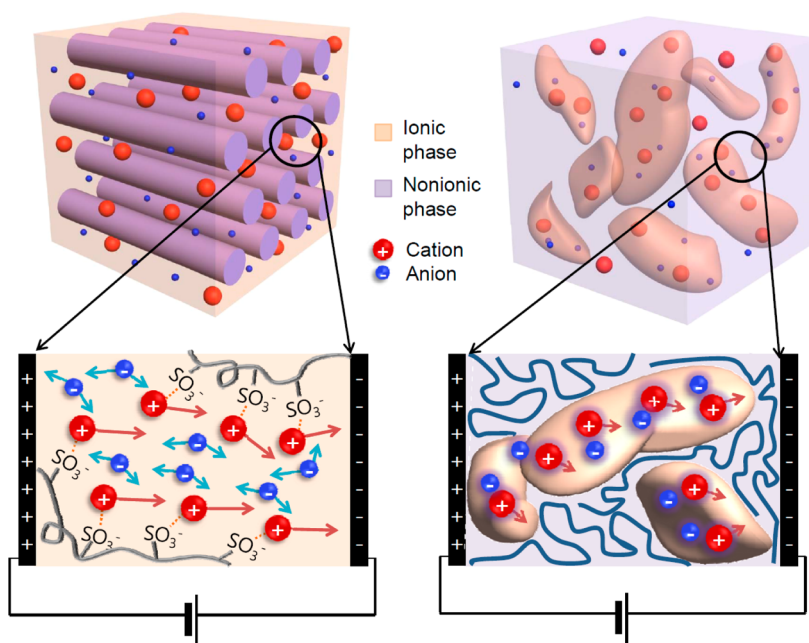


Figure 9. Schematic illustrations of the different ion migration behavior for T7(45) and PVdF(HFP) integrated with [HMIIm][TFSI]. By taking advantages of continuous PSS phases and highly interactive $-\text{SO}_3^-$ surfaces for sulfonated block copolymers, the effective IL dissociation and fast ion migration were facilitated under an applied electric field.

The conclusions deduced from the results of the *in situ* ATR and PGSE-NMR experiments are schematically illustrated in Figure 9. With [HMIIm][TFSI], the IL dissociation and cation/anion diffusion were facilitated by confining the IL within the PSS phases of the sulfonated block copolymers, leading to an efficient ion migration toward the oppositely charged electrode surface under a given electric field. This unique confinement is responsible for the unusual IL anion-dependent actuation performance observed in our actuators. Owing to the nonionic characteristics of PVdF(HFP), the cation and anion in [HMIIm][TFSI] are present in the form of ionic clusters, slowing the diffusion of ions. This contributed to a significant reduction in the bending strain for actuators made of IL-containing PVdF(HFP) compared to our actuators.

We conclude this paper by addressing the morphological response of the polymer layers to voltage stimuli. This allows us to identify the structural integrity of IL-containing polymers during actuation with valuable insights into mechanisms underlying improved actuation properties of the actuators composed of [HMIIm][TFSI] and sulfonated block copolymers. *In situ* SAXS experiments on the actuators were carried out with a function generator and an oscilloscope. To examine changes in the morphology of IL-containing polymers during actuation, an X-ray beam was transmitted along the thickness direction of the samples under applied voltages of ± 3 V. Three representative cross-sectional areas, i.e., near cathode, center, and near anode, were investigated with a reduced X-ray beam size of $30\ \mu\text{m}$ at the sample position, enabling accurate measurements of position-dependent morphologies beneath the electrodes. Scattering data were taken in each region after bending, as schematically illustrated in Figure 10a.

Representative SAXS profiles, measured under 3 V, are shown in Figure 10b for the T7(45) copolymer with 30 wt % [HMIIm][TFSI]. Nonsignificant alterations in the domain size with intact HEX morphology, as indicated by the series of Bragg peaks ($\blacktriangledown$) with the ratio of $1:\sqrt{3}:\sqrt{4}:\sqrt{7}$, were

observed in region II under applied voltages. This implies that the net movement of ions was offset at the center of the polymer layer. On the other hand, we found a significant disruption of the HEX morphology in region I (beneath the cathode), accompanied by an increase in the domain spacing of 1.02 nm. This denotes that the degree of phase separation was mitigated by steric repulsion in excessive [HMIIm] cations. In region III near the anode, a reduction in the domain size of 0.78 nm was determined, although the long-range order of the HEX structure was preserved. This signals that while the extraction of [HMIIm] cations from the PSS phases yielded the shrinkage of the structure, the accumulation of [TFSI] anions made less of an impact on the self-assembly of IL-containing polymers. These changes were entirely reversed by applying -3 V, which was reproducible under repeated voltage switching, suggesting that the durable motion of our actuators (Figure 6b) is attributed to the structural integrity of the IL-containing polymer layers.

We also investigated the morphological changes of [HMIIm]-[TFSI]-containing D17(42) and [HMIIm][PF₆]-integrated T7(45) copolymers under applied voltages to elucidate the effects of the block architecture and IL type on the morphological response of IL-containing polymers. Overall, qualitatively similar trends were seen for all samples in terms of domain size and degree of ordering. As summarized in Figure 10c, the [HMIIm][TFSI]-containing D17(42) copolymer displayed 1.40 nm increase and 1.08 nm decrease in the domain size near the cathode and anode, respectively. This corresponds to 5% peak-to-peak changes in lattice spacings with respect to the domain size of the sample without an applied electric field, which is slightly smaller than the 7% change obtained with [HMIIm][TFSI]-containing T7(45). Interestingly, as plotted in Figure 10c, the [HMIIm][PF₆]-containing T7(45) copolymer showed considerably smaller changes in the domain size (0.4 and -0.25 nm) in response to step voltages of ± 3 V compared to the [HMIIm][TFSI]-

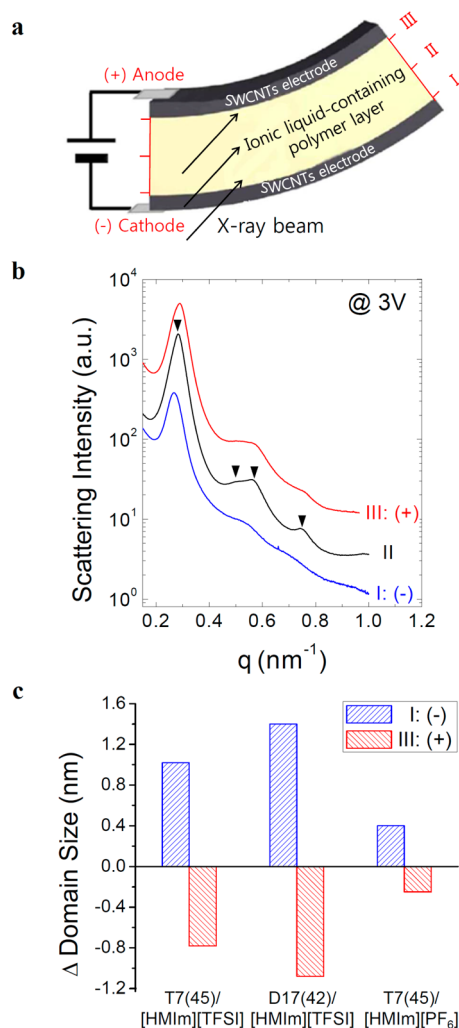


Figure 10. (a) Schematic drawing of the measurement setup of *in situ* SAXS experiments and (b) representative SAXS profiles obtained with the actuator comprising T7(45) and [HMIIm][TFSI], obtained in regions I, II, and III under 3 V. (c) Change in domain size of the HEX structure in regions I and III for [HMIIm][TFSI]-containing sulfonated block copolymers by varying polymer matrix, as indicated in the figure.

integrated analogue. This is in good agreement with the poor actuation performance of the actuators composed of [HMIIm]-[PF₆] and sulfonated block copolymers (Figure 3b).

This leads us to conclude that the actuation performance is intimately associated with the swelling and shrinkage behavior of IL-containing polymer layers. The observed actuator properties may be further controlled by varying the type of self-assembled morphology of IL-containing polymers, which will be the subject of our future studies. Our study clearly suggests the importance of the molecular design of polymer layers in actuators to improve the ion transport properties and mechanical integrity. Our work will contribute to the identification of key parameters to create synergetic molecular interactions in polymer electrolytes for the advancement of diverse electromechanical devices.

CONCLUSIONS

Key factors affecting the electromechanical properties of ionic polymer actuators were determined by employing a set of linear sulfonated di- and triblock copolymers and imidazolium ILs. The IL-containing sulfonated block copolymers displayed well-

defined hexagonal cylindrical morphologies where the IL was selectively incorporated into PSS phases. The PMB blocks located every tens of nanometers impart mechanical toughness during the actuation. The order of the actuation performance of our actuators based on IL-containing sulfonated block copolymers with three imidazolium ILs ([TFSI] > [BF₄] > [PF₆]) contrasted with the trend observed with IL-embedded PVdF(HFP) ([PF₆] > [BF₄] > [TFSI]). The use of sulfonated triblock copolymers in actuators enables us to achieve largely improved actuation performance by offering improved mechanical integrity even at large IL loadings. Remarkably, our actuators demonstrated 2–10-fold increases in the bending strain compared to those made with IL-embedded PVdF(HFP), signaling the importance of having well-defined ionic phases in the polymer matrix to improve the actuation performance by taking advantage of organized ion migration along the less tortuous ion-conduction pathways. PGSE-NMR experiments further demonstrated the facilitated ion diffusion within the sulfonated block copolymer matrix by manipulating molecular interactions between the IL and the polymer matrix by confining the IL in nanoscale PSS phases containing highly interactive –SO₃H surfaces.

A central question critical to actuation performance is how the mobile ions migrate and accumulate near charged electrodes. To determine this, we carried out in-depth studies on the actuation mechanisms, characterized directly by *in situ* ATR-FTIR and *in situ* X-ray scattering experiments under applied electric fields. While fast migration of both cations and anions, dominated by cation diffusion toward the oppositely charged electrode surface, was demonstrated for sulfonated block copolymers incorporated with [TFSI]-based ILs, it appears that for the same IL dispersed within the PVdF(HFP) matrix the strong Coulombic interactions between the IL cations and anions impede their separation under voltage stimuli, slowing the bending response of the actuators. Our results suggest that nanostructured ionic block copolymers with tailored mechanical strengths are an attractive alternative for new polymer matrices in ionic polymer actuators and their associated technologies.

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Notes

The authors declare no competing financial interest.

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